

BIBEK GUPTA

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Summary

Software & GenAI Engineer, B.Tech CSE at KIIT University (CGPA: 9.05/10), graduating 2027. IBM-certified AI/ML Engineer with production-deployed LLM systems spanning multi-agent orchestration, RAG pipelines, and VLM edge AI; strong foundation in data structures, algorithms, REST API design, and full-stack development.

Technical Skills

Programming Languages: Java, Python, JavaScript, PHP, C, SQL

Core CS & Backend: Data Structures & Algorithms, Object-Oriented Programming, MongoDB, OracleDB

Generative AI & LLMs: Fine-tuning, Prompt Engineering, LLM Evaluation, Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Orchestration, ReAct Agents, Chain-of-Thought Reasoning, Vision-Language Models (VLM), Small Language Models (SLM), Edge AI

LLM Frameworks & Libraries: LangChain, LangGraph, Hugging Face Transformers, OpenAI API, Groq, Llama 3, NVIDIA Nemotron, Pydantic, OpenRouter API

Vector Databases & Retrieval: ChromaDB, FAISS, Pinecone, Semantic Chunking, Embedding Models

AI/ML & Deep Learning: TensorFlow, PyTorch, Neural Network, Scikit-learn, Transformers, CNN, RNN, LSTM, NLP, Feature Engineering, Hyperparameter Tuning

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis, Exploratory Data Analysis (EDA), Power BI

Web Development & APIs: React, Node.js, FastAPI, Flask, Streamlit, RESTful APIs, Tailwind CSS

Cloud & Tools: Docker, Git, GitHub Actions, CI/CD, Railway, Cloudflare Pages, Render, AWS (Familiar)

Experience

Summer Intern – National Informatics Center (NIC), Govt. of India

May 2026 – Jul 2026

Shillong, Meghalaya, India

- **AMO ESIS Website Revamp** : improved responsiveness and stability of a live government platform serving 1000+ users, reducing 90% of reported load/UI issues; applied production software engineering practices in a regulated environment
- **Directorate of Housing Portal** – Updated and optimized the government web portal backend, resolving 40+ issues and coordinating third-party compliance audit submission with 100% first-pass approval with all mandatory GIGW 3.0 rules
- **Directorate of Home Guards and Civil Defence** – Developed an internal operations dashboard, automating 5+ manual workflows and reducing administrative processing time by 60%

Projects

AgentResearch.ai – Autonomous Multi-Agent Research Pipeline *Live*

Jan 2026 – Feb 2026

Python, LangGraph, LangChain, Llama 3.3 70B (Groq), Tavily Search, HuggingFace FLUX.1, FastAPI, React 18

- Architected a 5-node agentic AI orchestration pipeline using LangGraph with LLM-powered autonomous reasoning and Tavily-powered live web retrieval across three routing modes (closed_book, hybrid, open_book)
- Built and deployed a decoupled React 18 frontend + FastAPI backend with parallel LLM inference, REST API integration, HuggingFace FLUX.1 image generation, and real-time blog library management; live on Cloudflare Pages and Render

OcularSLM – Local Vision-to-Action Desktop Agent *Github*

Jun 2026 - Jul 2026

Python, NVIDIA Nemotron, ChromaDB, Pydantic, VLM, SLM, Edge AI, RAG

- Built a local Vision-to-Action desktop AI agent on edge hardware using NVIDIA Nemotron-Nano-12B-VL (VLM) for visual perception, Nemotron-Super-120B (SLM) as reasoning core, ChromaDB RAG for context retrieval, and Pydantic for structured function execution
- Achieved 95% reduction in screenshot payload (3MB to 150KB), 85% drop in prompt token overhead, and sub-15ms semantic retrieval latency – enabling real-time agentic desktop automation on local hardware

DocQueryAI – RAG-Powered Document Intelligence Platform *Live*

Dec 2025 – Jan 2026

Python, LangChain, Llama 3, ChromaDB, FAISS, Sentence Transformers, PyPDF2, Streamlit, Railway

- Designed and deployed production RAG pipeline using LangChain and Llama 3 with semantic chunking, ChromaDB vector retrieval, FAISS-compatible embedding storage, and sentence-transformer embedding models – achieving 90%+ context-aware QA accuracy on 50+ page documents, 99% uptime, sub-2-second response latency

Education

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar

Jul 2023 – Jun 2027

B.Tech in Computer Science and Engineering

CGPA: 9.05/10.0

Certifications and Achievements

Google Developer Groups Solution Challenge 2026 – Recognized for successfully designing, building and submitting a prototype solution utilizing artificial intelligence tools **(2026)**

IBM SkillsBuild AI/ML Professional Certificate – Edunet Foundation, 95% assessment score **(2025)**

Top 10% Ranking – MumbaiHacks, India's largest Agentic AI hackathon, 3000+ participants **(2025)**

UiPath Certified Automation Developer – Deployed 3 RPA workflows achieving 60% efficiency gain, KIIT Training & Placement Cell **(2025)**